

RESOURCE CONSTRAINTS

Model size

Qwen 1.7B · 4B · 8B · 9B

Quantization

fp16 · 8-bit · 4-bit · 2-bit

Turn budget

≤ 20 questions / round

Tool budget

≤ 5 USE_HINT · ≤ 5 WEB_SEARCH

⚡ Carbon tracker

kWh · CO₂ per game

configure agents

AGENTIC LOOP — orchestrator coordinates sub-agents & tools; repeats until guesser wins or turn = 20

GUESSER TEAM

ANALYST sub-agent

agent-as-tool · does not consume a turn

- reads transcript so far
- summarises evidence
- narrows candidate set
- suggests next info-gain split

RESEARCHER sub-agent

agent-as-tool · does not consume a turn

- searches public 20-Q transcripts
- extracts common strategies
- reports frequent secret words
- proposes high-yield questions

ORCHESTRATOR (Guesser)

size × bit-width

each turn, picks ONE action:

- CALL ANALYST(prompt) *agent-as-tool*
- CALL RESEARCHER(prompt) *agent-as-tool*
- ~~USE_HINT / WEB_SEARCH: {q}~~
- ~~QUESTION: yes/no question~~

USE_HINT → next pre-built hint (≤ 5)

WEB_SEARCH: {q} → DuckDuckGo snippet (≤ 5)

SECRET KEEPER LLM

system prompt holds the secret

one-word reply:

- YES / NO (to a question)
- CORRECT / WRONG (to a guess)

QUESTION / GUESS

YES / NO | CORRECT / WRONG

↺ next turn

GameRecord (transcript · turns · sub-agent calls · tool counts)

POST-GAME EVALUATION (excluded from carbon clock)

Layer 1 — Game outcome

Did the guesser identify the secret, and how efficiently?

Layer 2 — Question quality

How well-targeted are the guesser's questions?

Layer 3 — Win verifier

Independent embedding-based check that the final guess truly matches the secret.